Core Finance
Week 10

MGMP 543

Prof. Benedict Guttman-Kenney

Spring 2025

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Jones Graduate School of Business

Hello! Today's Topics

- Week 9 Recap
- Quiz 8
- Week 9 Examples
- New Topic: Options & Derivatives
- Week 10 Examples

Week 9 Recap

Why Do We Need to Adjust Betas for Conglomerates?

- Boeing’s commercial and government business arms carries different risks.
- We want a rate of return that reflects the risk of the commercial arm.

Why Do We Need to Unlever Betas to calculate a Commercial Beta?

$$\beta_{Unlevered} = \beta_U = \frac{\beta_L}{[1 + (1 - T_C)\frac{D}{E}]}$$

- We want to compare like-for-like.
- Need to adjust betas to account for firms having different capital structures.

From Exhibit 10…	Boeing	Lockheed Martin	Northrop Grumman
Debt/Equity	0.525	0.410	0.640
Levered Beta Given (S&P500, 60 months)	0.80	0.36	0.34
Unlevered Beta Calculated (S&P500, 60 months)	0.596	0.284	0.240

Why Do We Need to Relever Betas?

- Levered betas are used in WACC!
- The cost of capital depends on a firm’s capital structure.

What Did We Learn Last Week?

Quiz 8

Q2. How much money will shareholders have in three years if Benedict pays out now and shareholders invest in treasuries for three years? Round your answer to the nearest dollar.

GK firm has $2,000 of extra cash. Benedict, the CFO, is thinking about what to do with the money. He can put it into treasuries for three years currently yielding 6% or pay it out to shareholders now. Of course, shareholders themselves also have access to the treasury market. Most of the equity investors in GK are poor and pay only a 15% personal tax rate. The tax rate on dividends is 18%. The corporate tax rate is 32%.

Q3. How much money will shareholders have in three years if Benedict invests in treasuries and pays out in three years? Round your answer to the nearest dollar.

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Q4. After how many years should Benedict pay out if he wants to do what's best for shareholders?

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Week 9 Examples

Uncertainty

Uncertainty in DCF Calculations (also applies to NPV)

Future is not known. There is uncertainty!

Should we change the cashflows or the discount rate or both?

- Don't update the discount rate as this *already* reflects the risks
- Update the cashflows: (i) probability weighting (ii) sensitivity analysis (changing assumptions) (iii) scenario analysis (iv) real options (later today)

Example of Probability Weighting in PV

Assume a discount rate of 10%.

$750 investment cost today.

Cashflows in perpetuity.

Possible Cash Flow	Probability
0	0.10
80	0.25
100	0.45
120	0.25

$$NPV = -\$750 + \frac{(\$0 \times 0.10) + (\$80 \times 0.25) + (\$100 \times 0.45) + (\$120 \times 0.25)}{0.1}$$

$$NPV = -\$750 + \$900 = \$150$$

Real Life Example of Scenarios

Exhibit 99.2

UNITED

A STAR ALLIANCE MEMBER

Investor Update[1] **Issue Date: April 15, 2025**

This Investor Update provides guidance and certain other forward-looking statements about United Airlines Holdings, Inc. (the "Company" or "UAL"). The information in this Investor Update contains the preliminary financial and operational outlook for the Company for second-quarter and full-year 2025, among other items.

The Company's outlook is dependent on the macro environment which the Company believes is impossible to predict this year with any degree of confidence. Accordingly, two guidance benchmarks are provided below. We undertake no obligation to update this guidance as circumstances evolve.

	Estimated 2Q 2025	Estimated FY 2025: Stable Environment	Estimated FY 2025: Recessionary Environment
Adjusted diluted earnings per share ("EPS")[2]	$3.25 - $4.25	$11.50 - $13.50	$7.00 - $9.00
Adjusted total capital expenditures (in billions)[3]	-	<$6.5B	<$6.5B

Adjusted Diluted EPS Supplemental Information:
The Company's guidance is based on consensus market macroeconomic expectations. However, a single consensus no longer exists, and therefore the Company's expectation has become bimodal – either the U.S. economy will remain weaker but stable, or the U.S. may enter into a recession. The Company is therefore providing two separate guidance benchmarks based on these two different macroeconomic views:

Stable Environment Scenario: While the Company is closely monitoring bookings, as of the date of this Investor Update, booking trends have been stable. Therefore, if demand remains consistent with the weaker, but stabilized demand trends from the six weeks prior to this Investor Update, along with the current forward fuel curve, the Company continues to expect to be within the initial guidance range of $11.50 to $13.50 in adjusted diluted earnings per share.

Recessionary Environment Scenario: If the U.S. economy enters a recession, we are modeling an incremental 5 point reduction in our total operating revenue in 2Q 2025 through 4Q 2025 compared to the stable environment scenario above. With that decrease in expected revenue, along with further capacity reductions from United but assuming no further cost relief from fuel, the Company expects adjusted diluted earnings per share of $7.00 to $9.00. For modeling purposes, each incremental 5 point reduction in revenue for the rest of the year would equal approximately $4.50 in adjusted diluted earnings per share if there were no further fuel or capacity offsets.

Real Life Example of Valuation: Bill Ackman / Hertz

Bill Ackman is CEO/Founder of Hedge Fund (Pershing Square Capital Management)

Tweeted last week revealing now own 20% of Hertz https://x.com/BillAckman/status/1912955549542744462

- “While Hertz has a highly leveraged balance sheet today, the substantial majority of the company’s debt does not mature until 2028 and 2029. Hertz has ample liquidity – including cash on hand, available revolver capacity, and debt capacity within its asset-backed securitization facilities – to support its fleet rotation, and to address maturities and other financial liabilities.”

- “Hertz is uniquely well-positioned in the current tariff environment, where auto tariffs are likely to cause used car prices to rise. Hertz owns a fleet of over 500,000 vehicles valued at approximately $12 billion. A 10% increase in used car prices would equate to a $1.2 billion gain on its auto assets – equivalent to approximately half of the company’s current market capitalization.”

- “At 7.5 times EBITDA, a valuation that we believe to be conservative in light of improvements in the competitive posture of the industry, we estimate that Hertz will be worth ~$30 per share by 2029. While the tariff announcements have created a near-term cloud over the travel industry – we have low expectations for Hertz’s Q1 and r first half results – we believe that over the intermediate term, the company will generate sustainably higher profitability.”

- “At 7.5 times EBITDA, a valuation that we believe to be conservative in light of improvements in the competitive posture of the industry, we estimate that Hertz will be worth ~$30 per share by 2029."

Week 10: Options & Derivatives

Real Options

Option Value?

DCF vs. Real Options

- **DCF** assumes that investment decision is “passive”
  - You decide whether or not to go ahead with a project.
  - If you go ahead, you see the project through, and any other decisions are separate.

DCF vs. Real Options

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  - If a project is going well, you may expand it
  - If a project is going badly, you may downsize it or close it down

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  - If a project is going well, you may expand it
  - If a project is going badly, you may downsize it or close it down
- **Real Options:** the ability (but not obligation) to modify projects
  - Option to expand (if higher demand higher than expected), abandon (if lower demand higher than expected), or delay (demand trending up but slower than expected)
  - Quantifiable
  - Will only use the option if it is beneficial to do so, otherwise do not use the option
  - Market Value = NPV + Option Value
- Projects that allow for ***more*** flexibility are ***more*** valuable, *especially* if there is more uncertainty in the forecast

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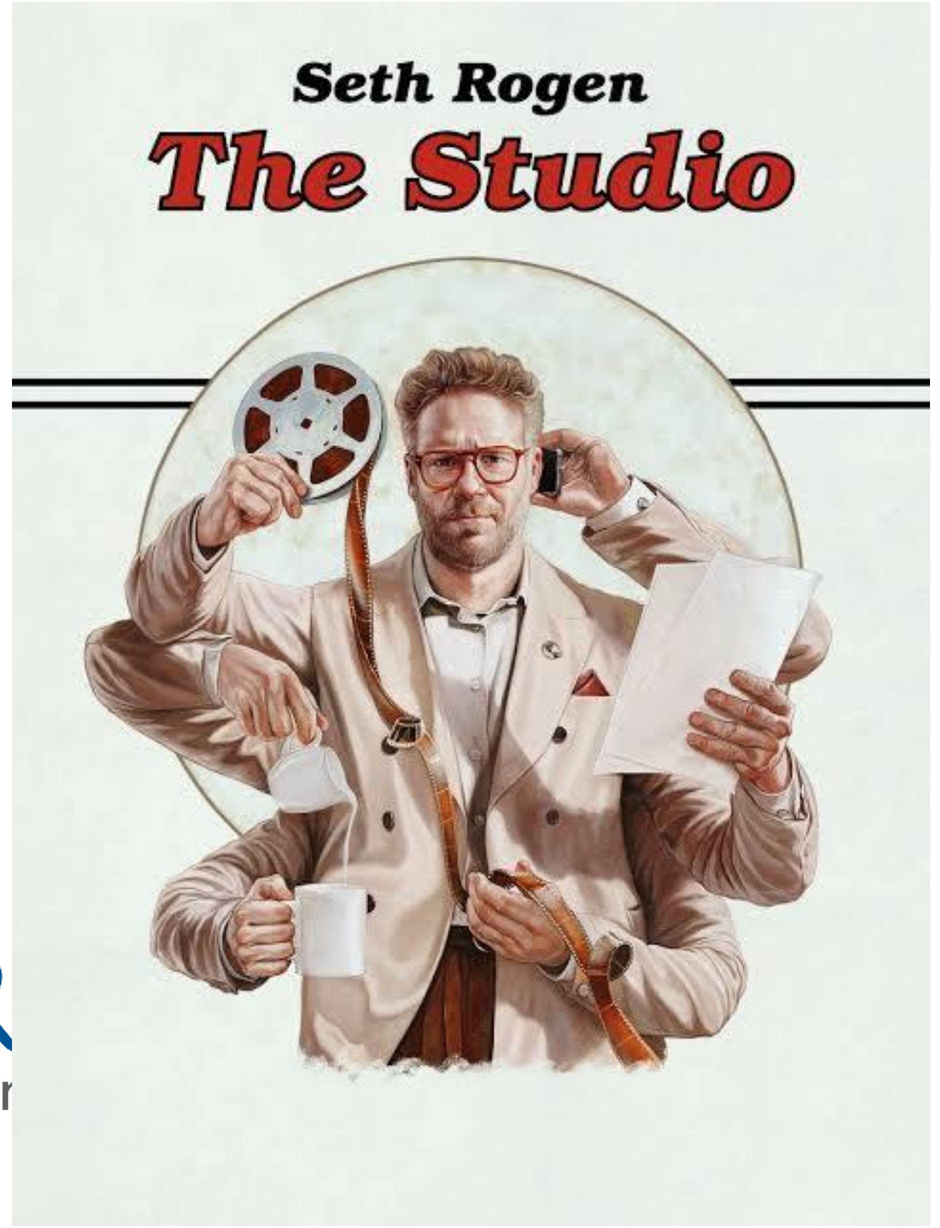
  - Market Value = NPV + Option Value
- Projects that allow for ***more*** flexibility are ***more*** valuable, *especially* if there is more uncertainty in the forecast
- "This may look like a bad business, but it gives us options to get into better business" No! This isn't a real option! A bad project (NPV < 0) is still a bad project!

Real Option Examples

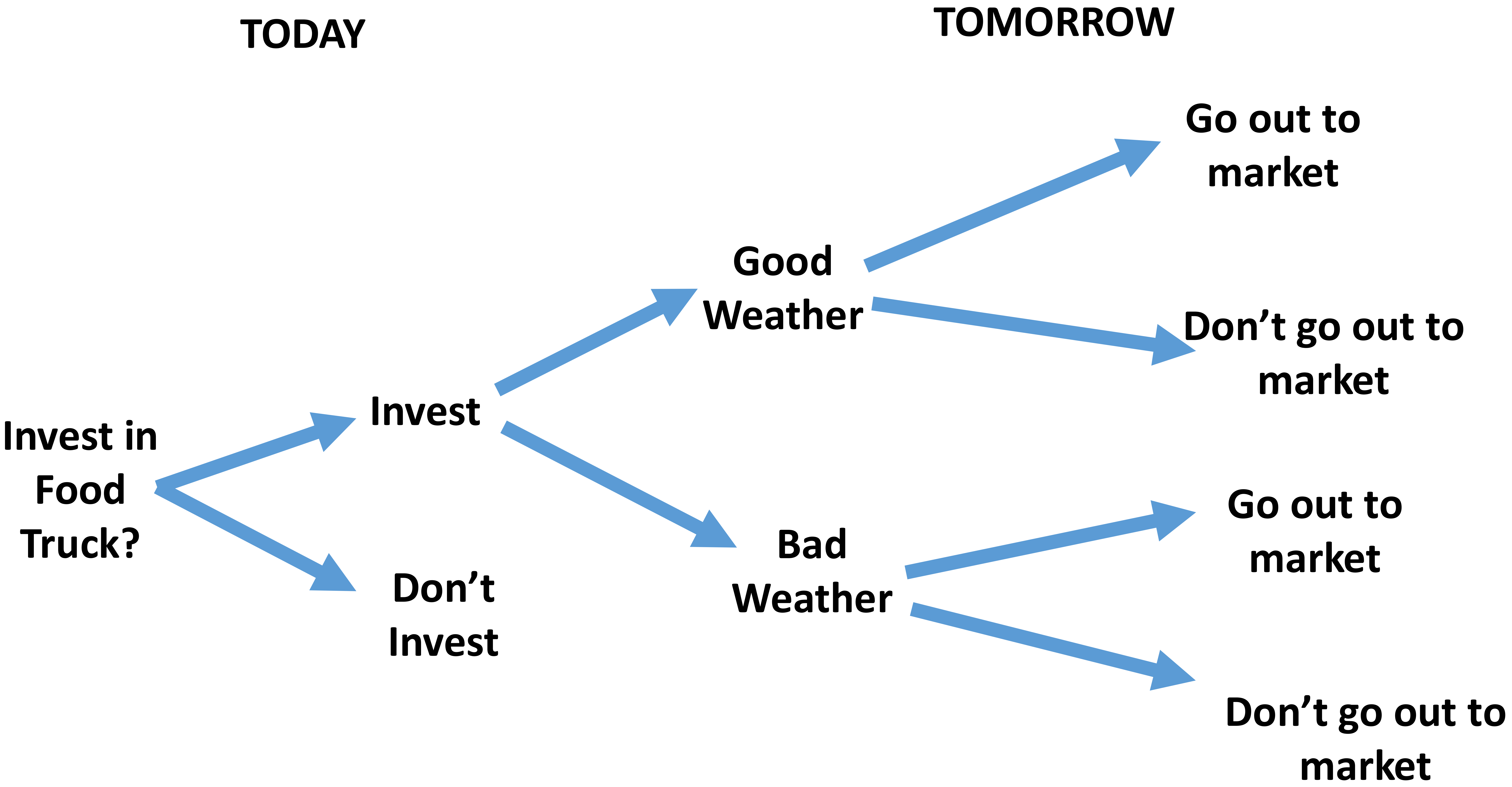
- **Pharmaceuticals:** Multiple stages of drug development (Pre-Clinical Trials, Phase I,II,III,IV) and at each stage company can decide whether to continue or halt project.
- **Movies:** A film studio may acquire the rights to produce sequels for a movie even before the original movie is released. If the movie does badly, no sequel. If the movie does well, produce the sequel.



Real Option Example

- GKadburys is a U.K. company of chocolate stores
- GKadburys is looking to expand its U.K. company into the U.S. (Austin & Houston)
- GKadburys can open stores in 2 cities ***simultaneously*** OR enter each city ***sequentially***
- GKadburys is uncertain on U.S. demand for its chocolate stores.
- Sequential entry is useful. Why?
  - GKadburys can evaluate the demand in Austin, and ***then*** management can use this information to decide whether to enter Houston.

Real Options in a Decision Tree



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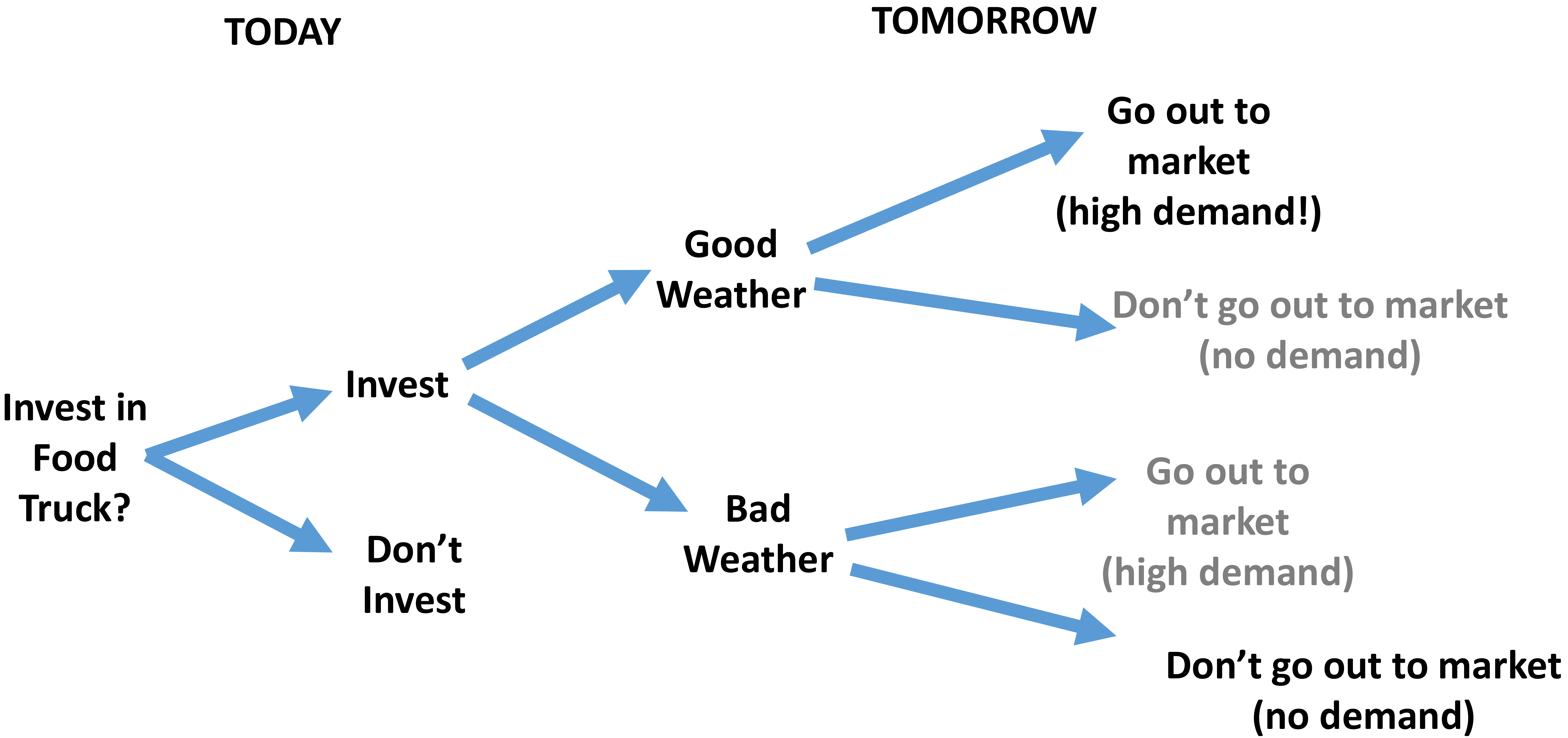

Options / Derivatives

Options are Everywhere!

- Options are used for investment decisions (expand/contract/abandon/delay projects)
- Many companies use options to hedge/manage their risks
- Many companies give their employees Stock Options

Jargon-Busting Financial Options

- An **Option** is a contract giving its owner the right but not the obligation to buy or sell an asset at a fixed price on or before a given date. The writer of this contract is required to buy or sell an asset.

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- **Strike Price / Exercise Price:** Fixed price in the option contract at which the holder can buy/sell the underlying asset.
- **Expiration Date:** Maturity date of the option. After this date, the option no longer exists.
    - Monthly Contracts (the most traditional form of options) usually expire at the end of the third Friday of each month.
    - Weekly Options (“Weeklys”) expire at the end of each trading week.
    - Zero-day to expiration options (“0DTE” / “Dailys”) expire at the end of each trading day.
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- **Naked/Uncovered Option:** Writer of contract does not hold the underlying asset (very risky strategy, need margin account)
- **Synthetic Options:** Replicate the trading strategy (payoffs) of options without directly owning options.
- We will go through two types of options: (i) **Call** Options and (ii) **Put** Options

Jargon-Busting Financial Options: An Example

- You are considering buying a $1 million property in Rice Village.
- For $1,000 the buyer sells you the right to buy that property at any time before 1st December 2025.
- **Strike Price / Exercise Price:** $1 million
- **Expiration Date:** 1st December 2025
- **Exercising the Option:** Exercise if fair market price is >$1 million
- **American Option or European Option?** American Option

Call Option

Call Me, May Buy

- A **call option** gives the owner the right to buy an asset at a fixed price during a particular time period.

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Call Me, May Buy

- A **call option** gives the owner the right to buy an asset at a fixed price during a particular time period.
- So, if the stock goes up, call me (and use the option) to buy!
- Another way to remember: If the stock price goes down, I won't (use my) call.

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- So, if the stock goes up, call me (and use the option) to buy!
- Another way to remember: If the stock price goes down, I won't (use my) call.
- Buying a call ("Long Call") is a "bullish" strategy because you want the price to increase. If the price increases, then the option becomes more valuable. (whereas "Short Call": selling a call option is more valuable if price increases)

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Call Option Example

Call option on IBM stock enables buying 100 IBM shares on 25 April at a price of $100.

Value of the (European) call option (C) <- value to the buyer/owner

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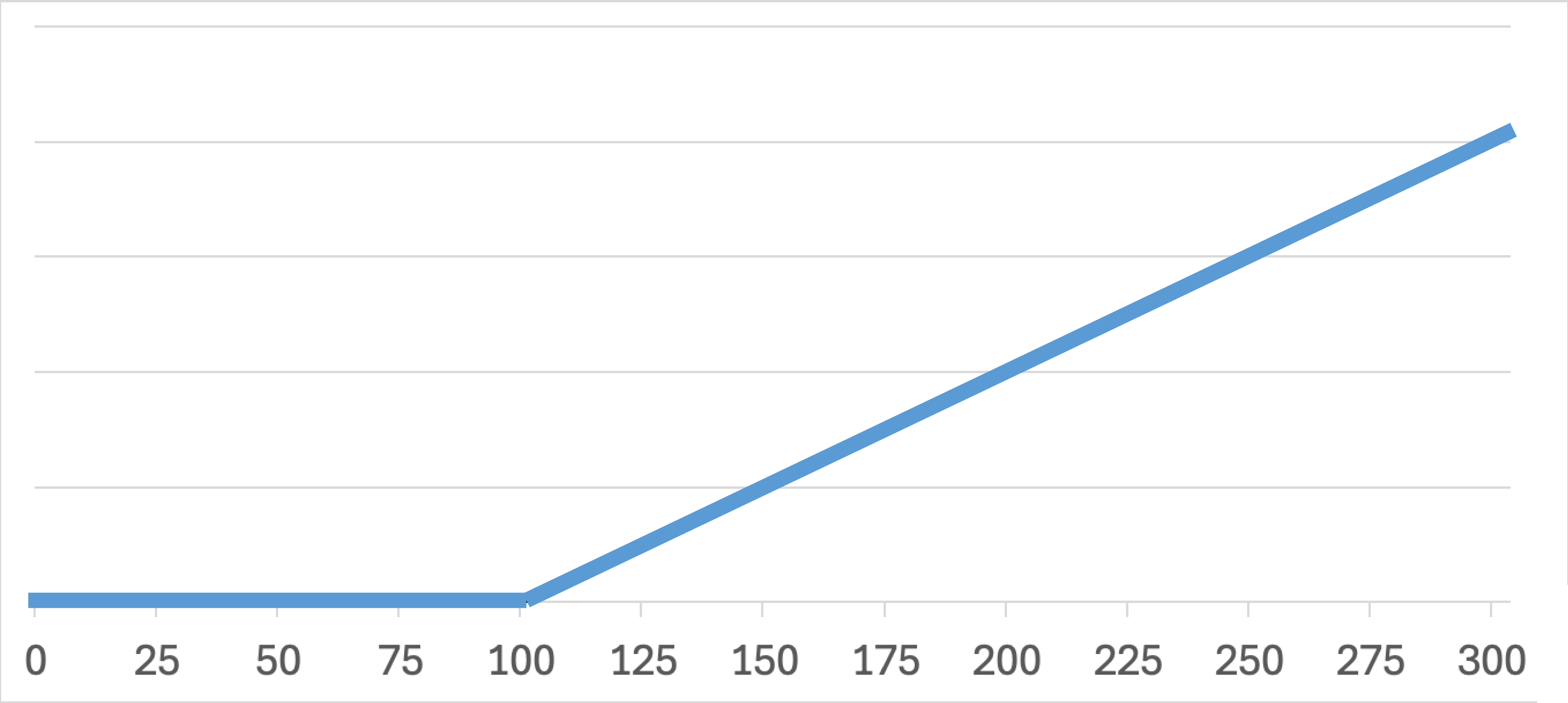
Value of the (European) call option (C) <- value to the buyer/owner

- Worth $0, if IBM stock price is <$100 on 25 April. “Out of the money”
- Worth IBM Stock Price – $100, if IBM stock price is $100+ on 25 April. “In the money”
- $C = \max(S_T - K, 0)$, where S_T is the stock price at expiration and K is the “strike price”
- Call option is more valuable if S_T ↑ or K ↓

Value of call option at expiration ($)



Value of common stock at expiration ($)

Put Option

High Prices **Put** Me Off Selling

- A **put option** gives the owner the right to sell an asset at a fixed price during a particular time period.
- So, if the stock goes up, put me off selling (using the option)!
- Another way to remember: If the stock goes down, I will (use my) put.
- Buying a put ("Long Put") is a "bearish" strategy because you want the price to decrease. If the price decreases, then the option becomes more valuable. (Whereas "Short Put": selling a put option is more valuable if price increases)

Put Option Example

Put option on IBM stock enables selling 100 IBM shares on 25 April at a price of $100.

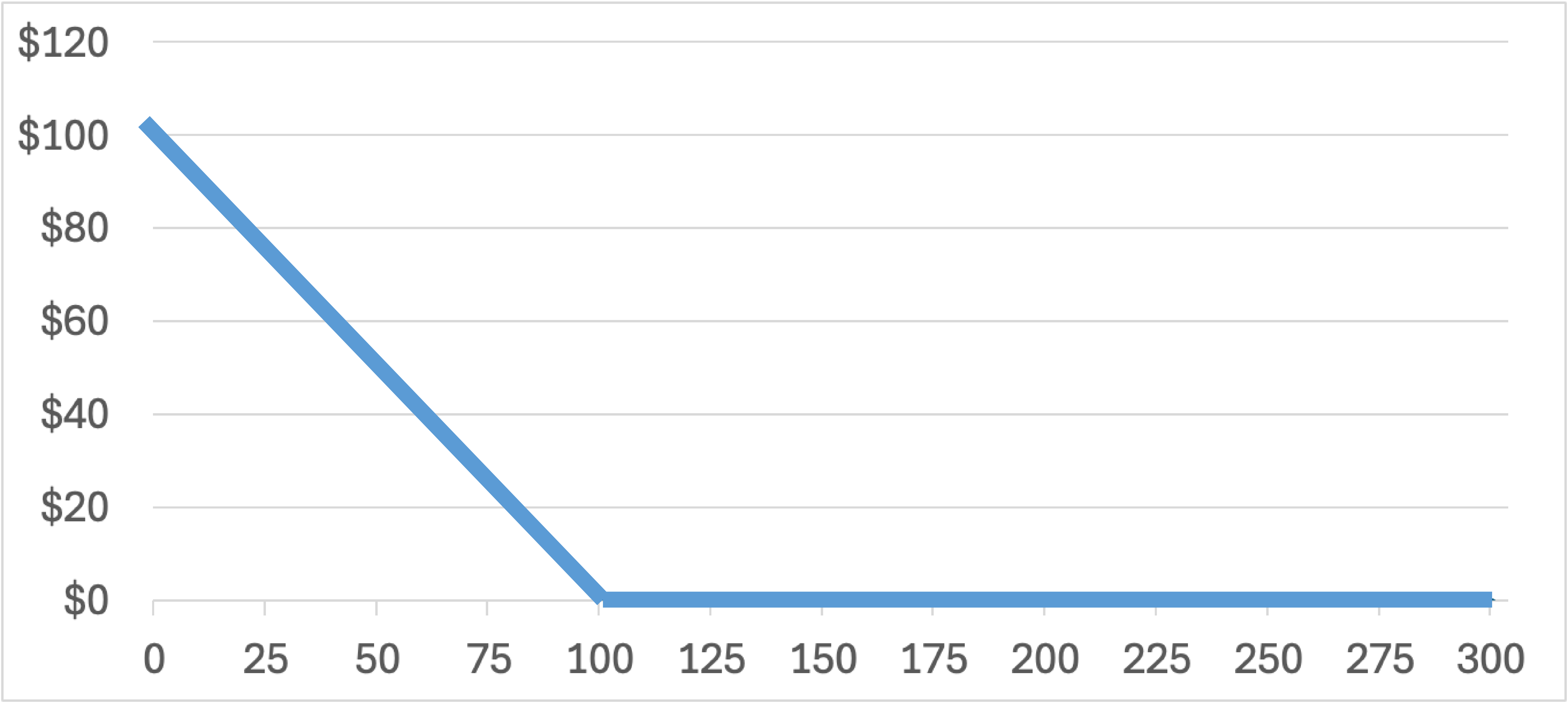
Value of the (European) put option (P) <- value to the buyer/owner

Put Option Example

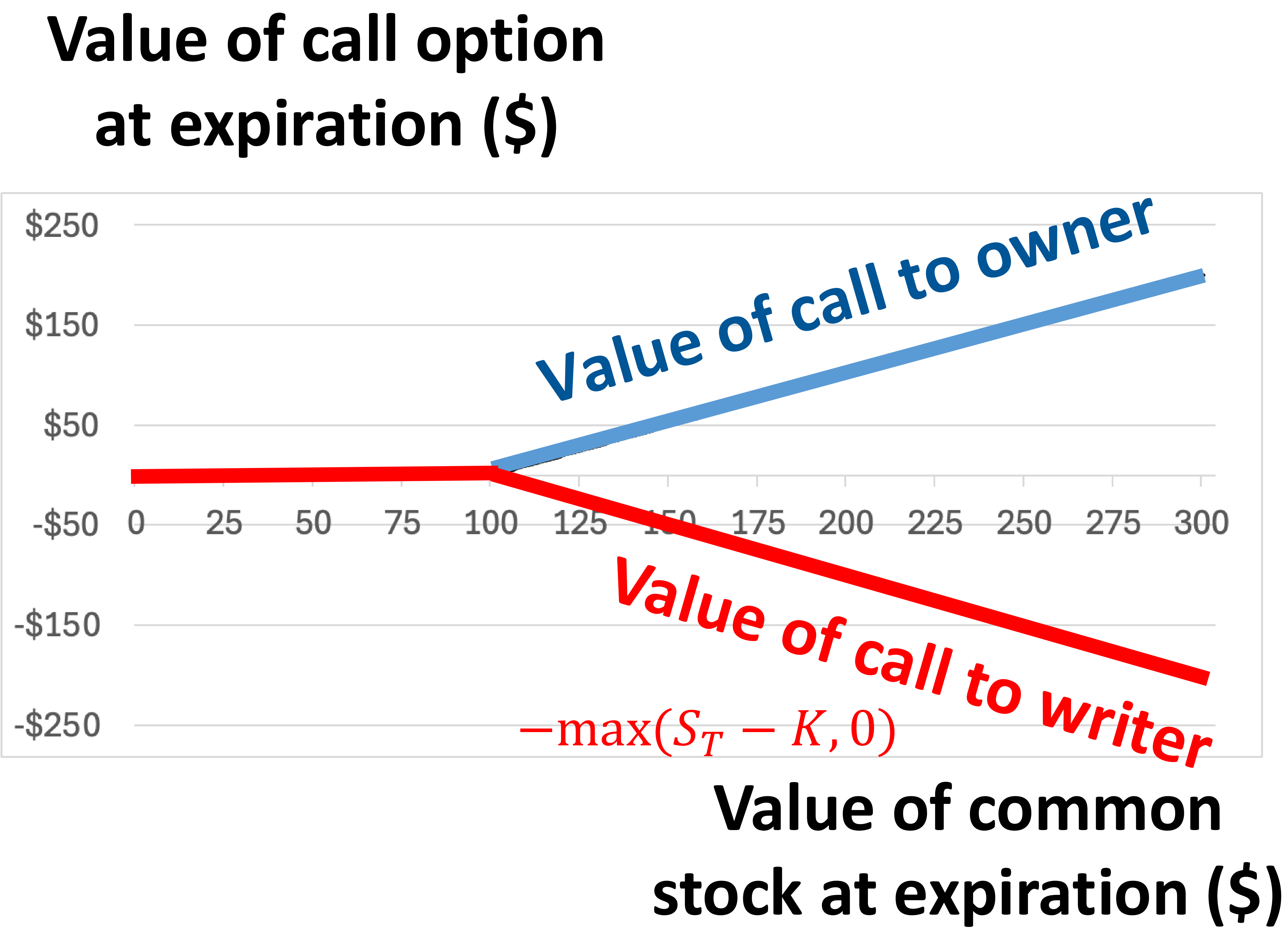
Put option on IBM stock enables selling 100 IBM shares on 25 April at a price of $100.

Value of the (European) put option (P) <- value to the buyer/owner

- Worth $0, if IBM stock price is $100+ on 25 April. “Out of the money”
- Worth $100 – IBM Stock Price, if IBM stock price is <$100 on 25 April. “In the money”
- $P = \max(K - S_T, 0)$, where S_T is the stock price at expiration and K is the “strike price”
- Put option is more valuable if S_T ↓ or K ↑



Payoffs for Writers/Sellers of Options



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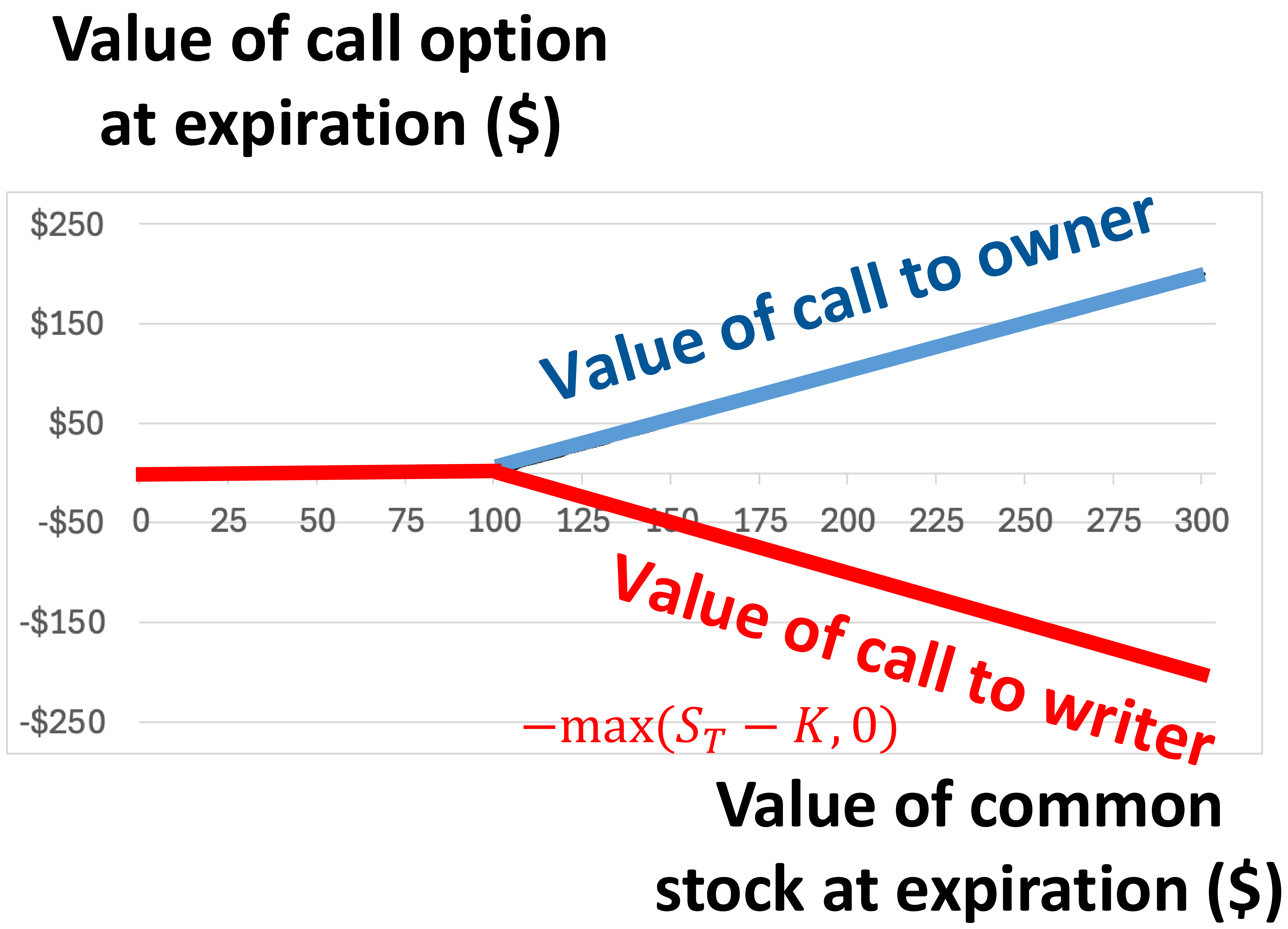

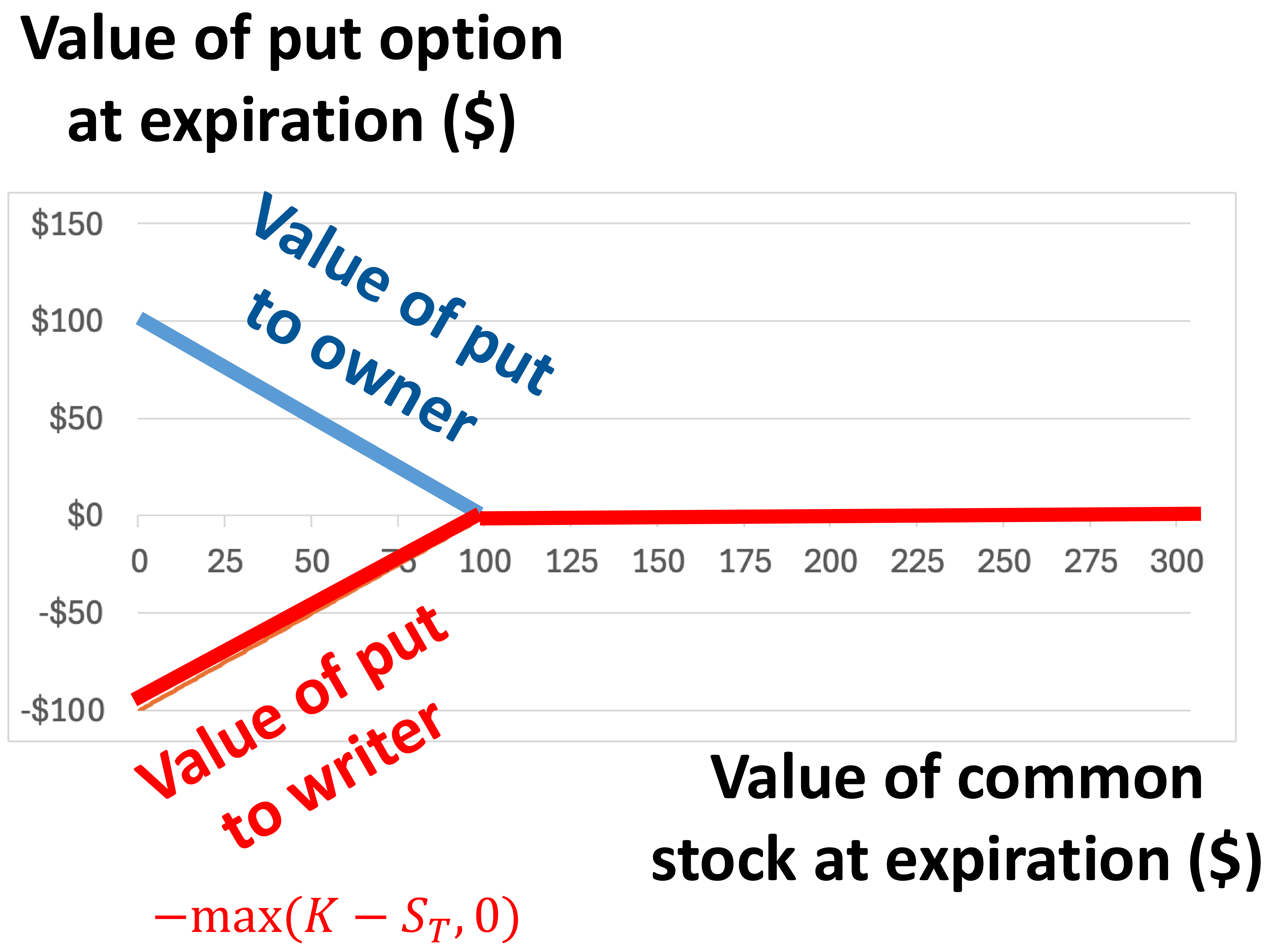

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Combining Options

What affects the value of Calls and Puts?

Increase in…	Value of Call Option	Value of Put Option
Value of underlying stock	+	-
Strike price	-	+
Volatility of underlying stock	+ (more volatility more upside)	+
Interest rate	+ (reduce present value of strike price, cheaper to buy later)	-
Time to expiration	+ (more time, more expected volatility)	+ (-) (more time, more expected volatility. But if stock price far below strike price then not)

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Combining Stocks/Bonds(/Other Securities) and Options (Calls/Puts) Produces Different Investment Strategies

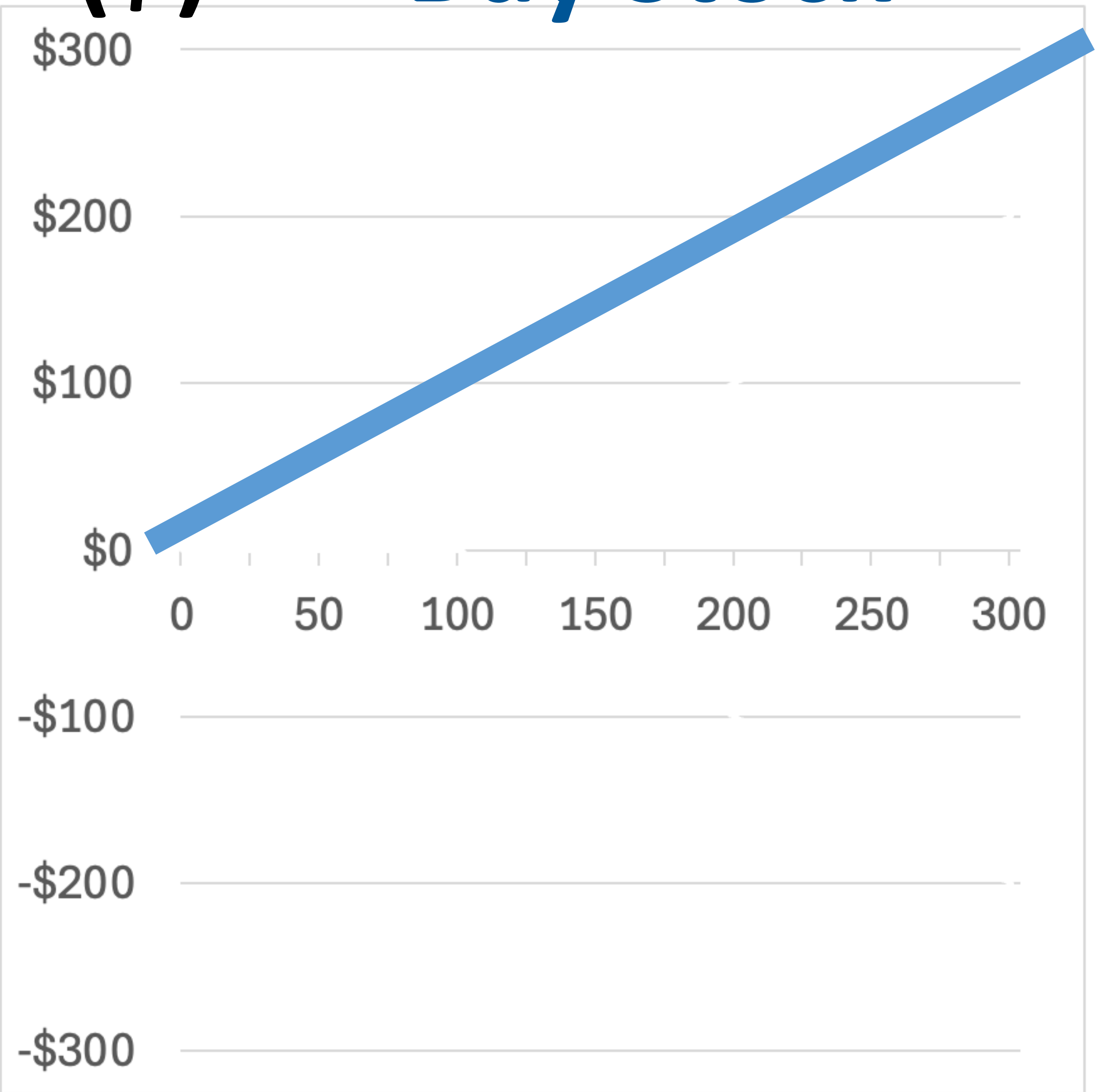
Aim: Protect downside (risk that stock falls in price)

Approach 1: Stock + Put

Total Payoff (Stock + Put)

Payoff ($)

Buy Stock



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Buy Put

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Value of common stock at expiration ($)

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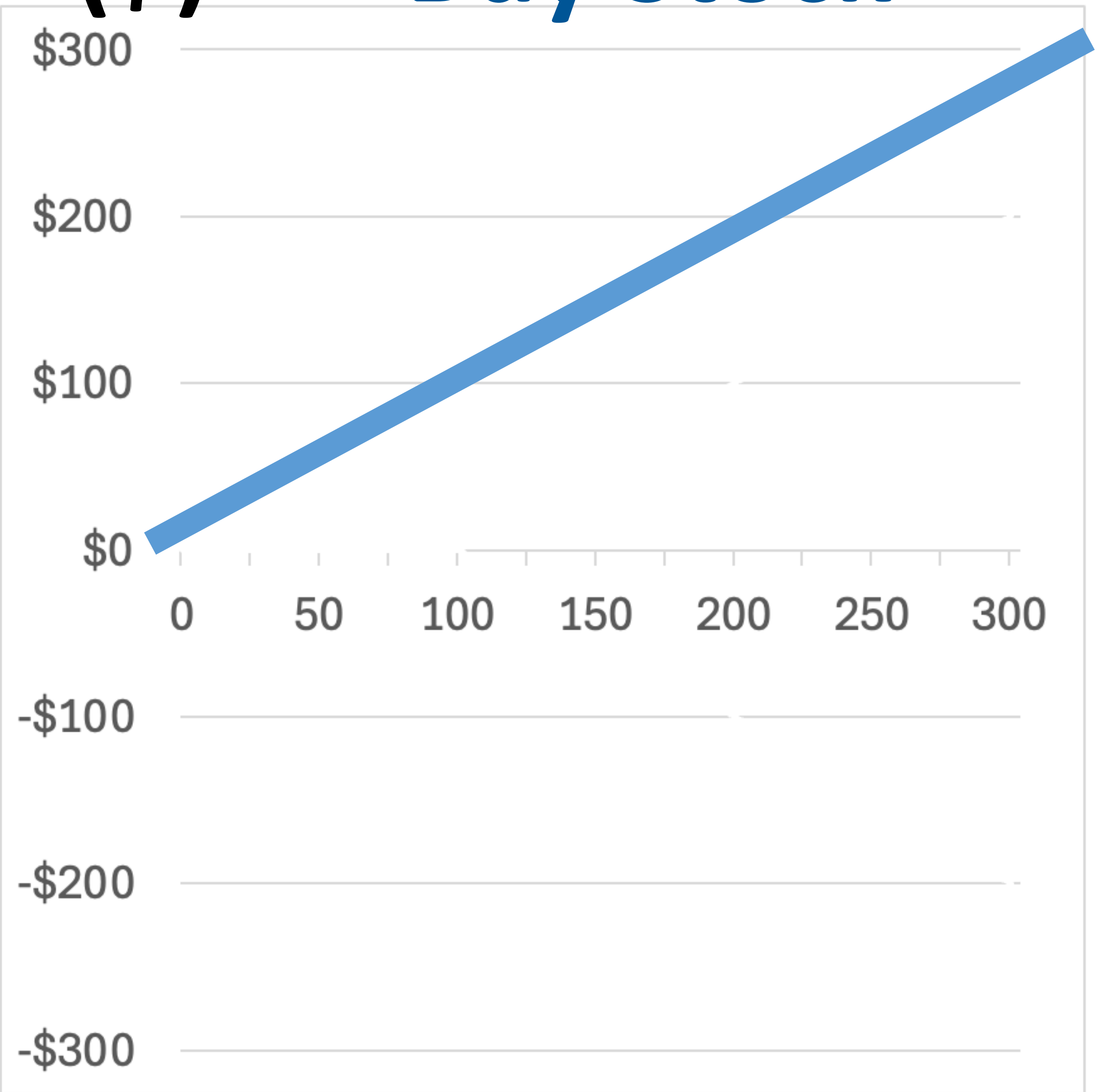

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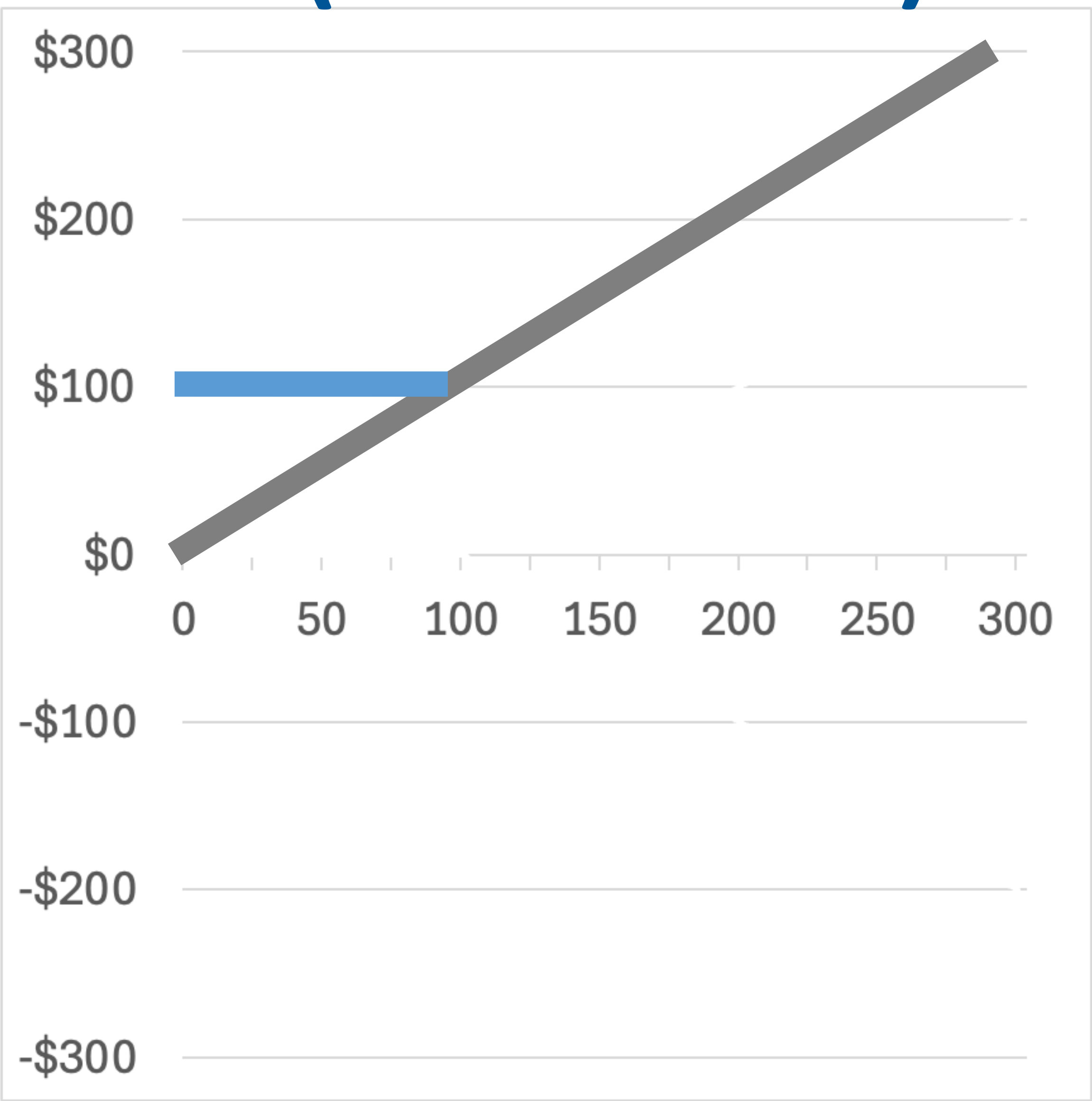


Value of common stock at expiration ($)

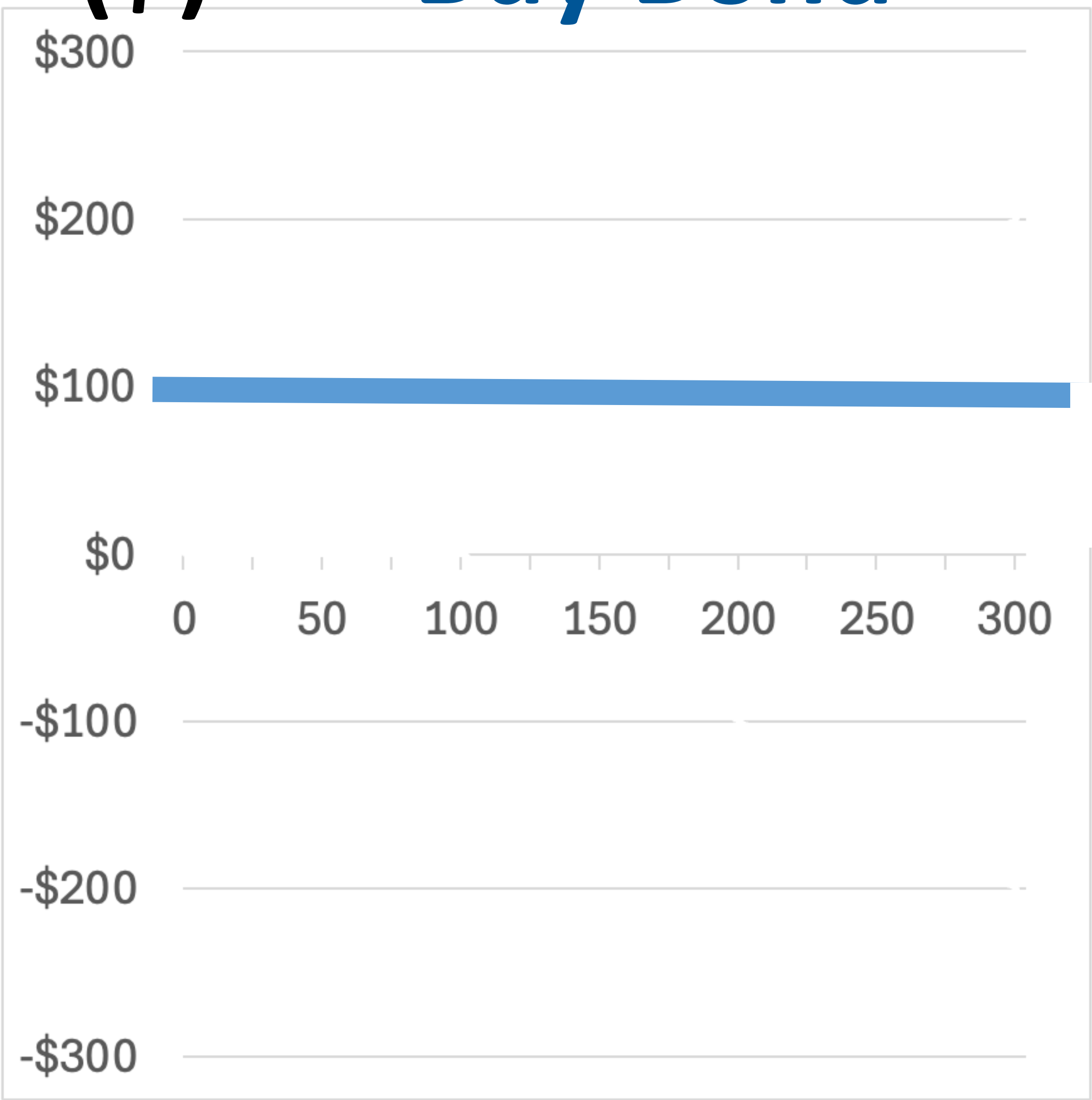
Combining Stocks/Bonds(/Other Securities) and Options (Calls/Puts) Produces Different Investment Strategies

Aim: Protect downside (risk that stock falls in price)

Approach 2: Bond + Call

Payoff ($)

Buy Bond



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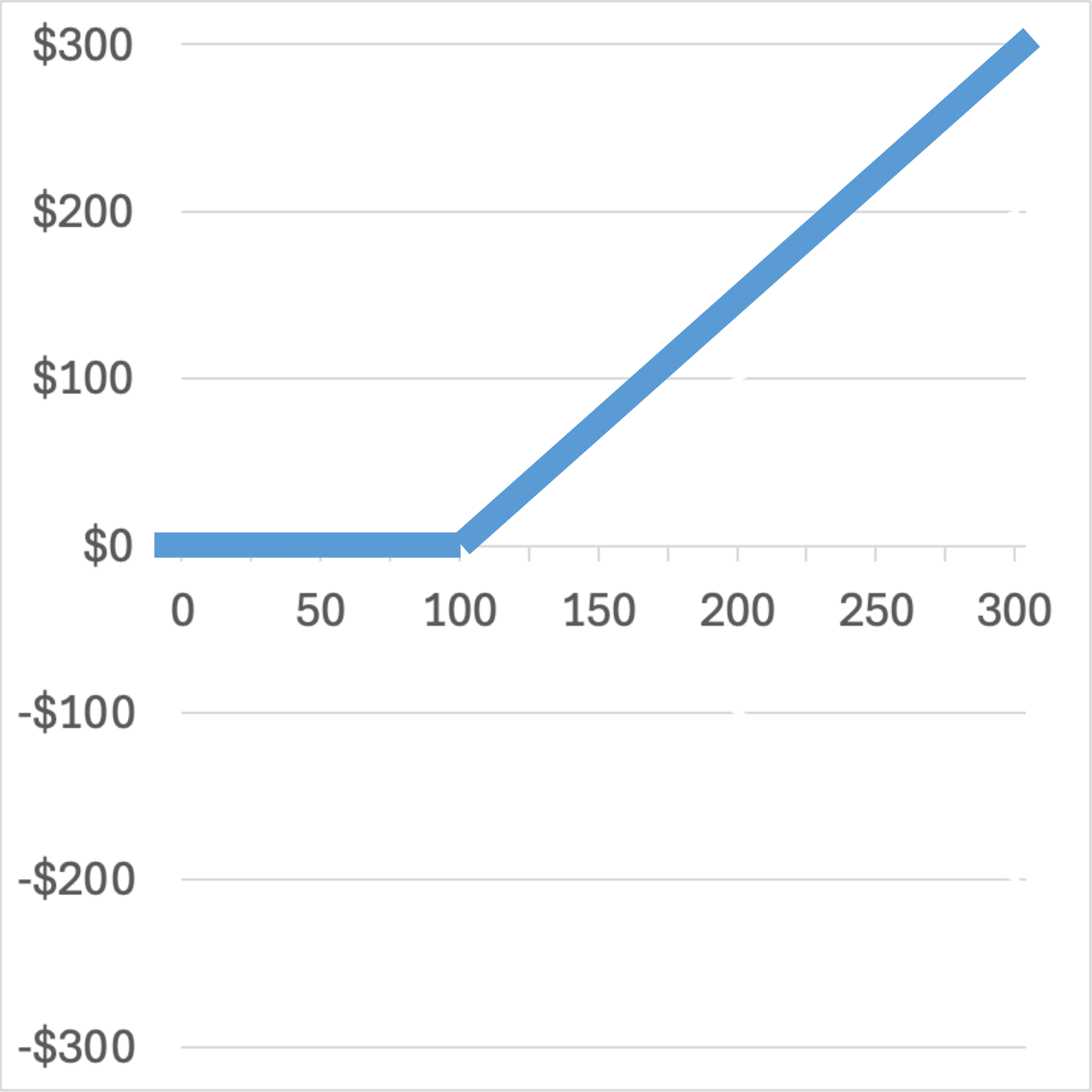
Buy Call



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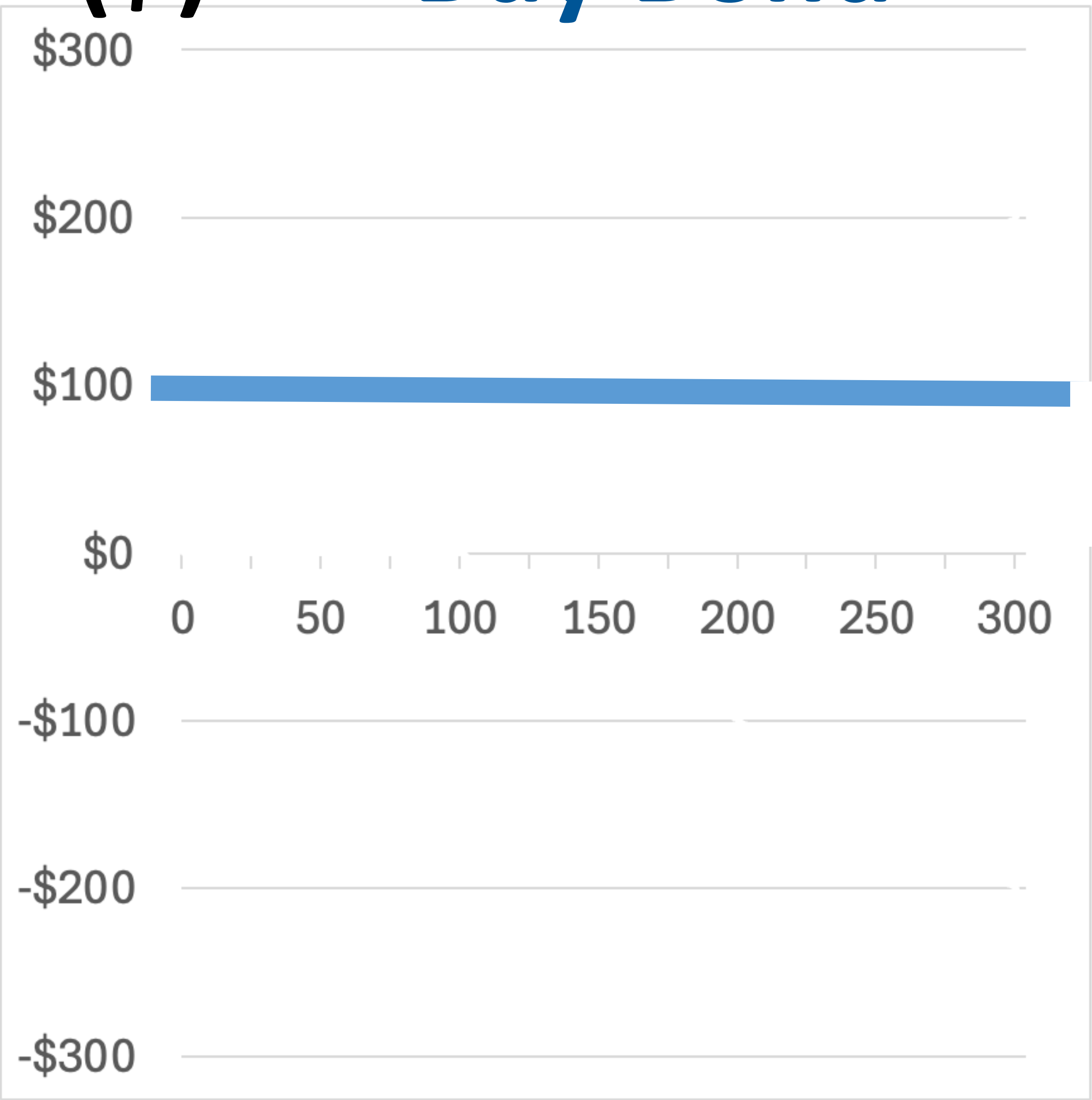

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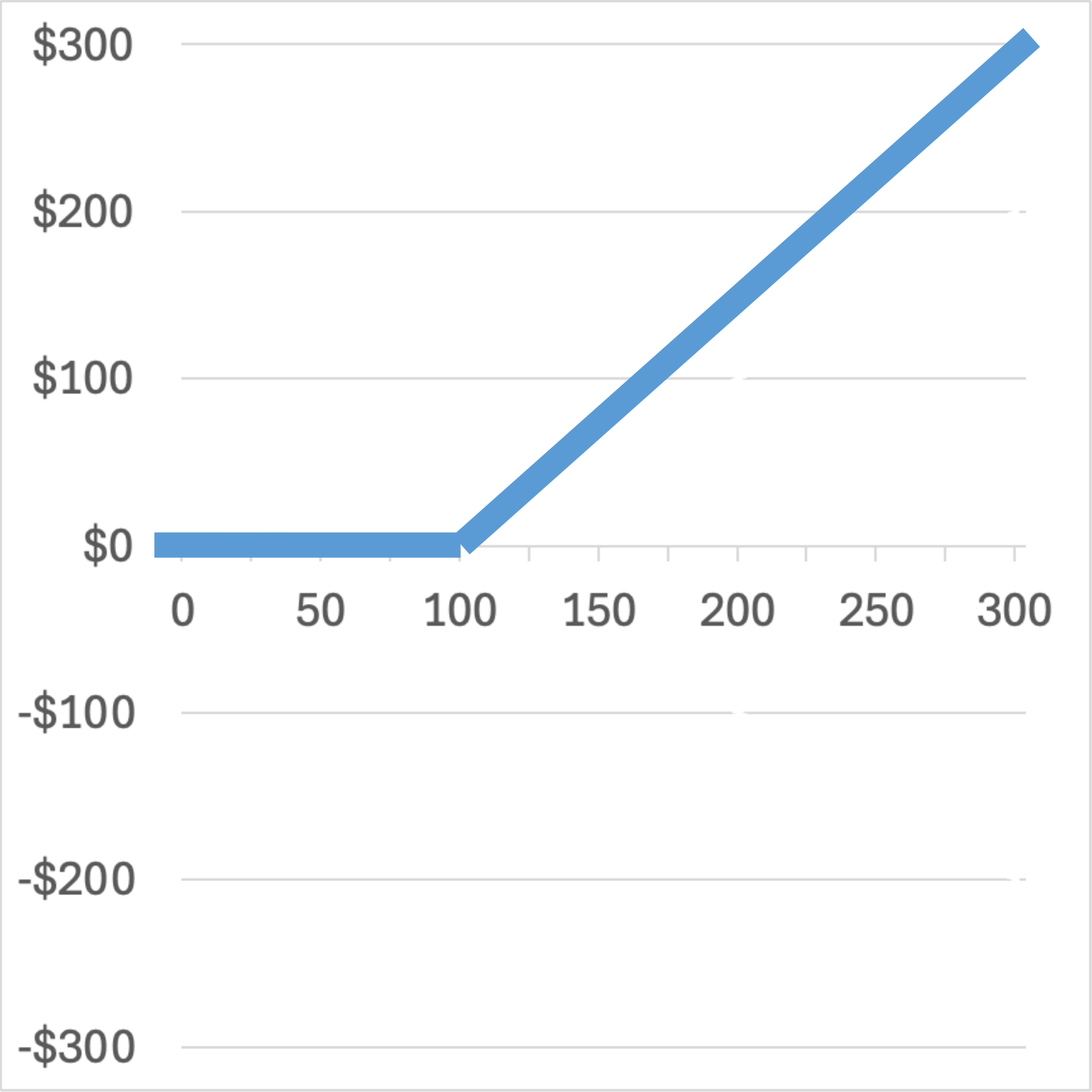
Buy Call



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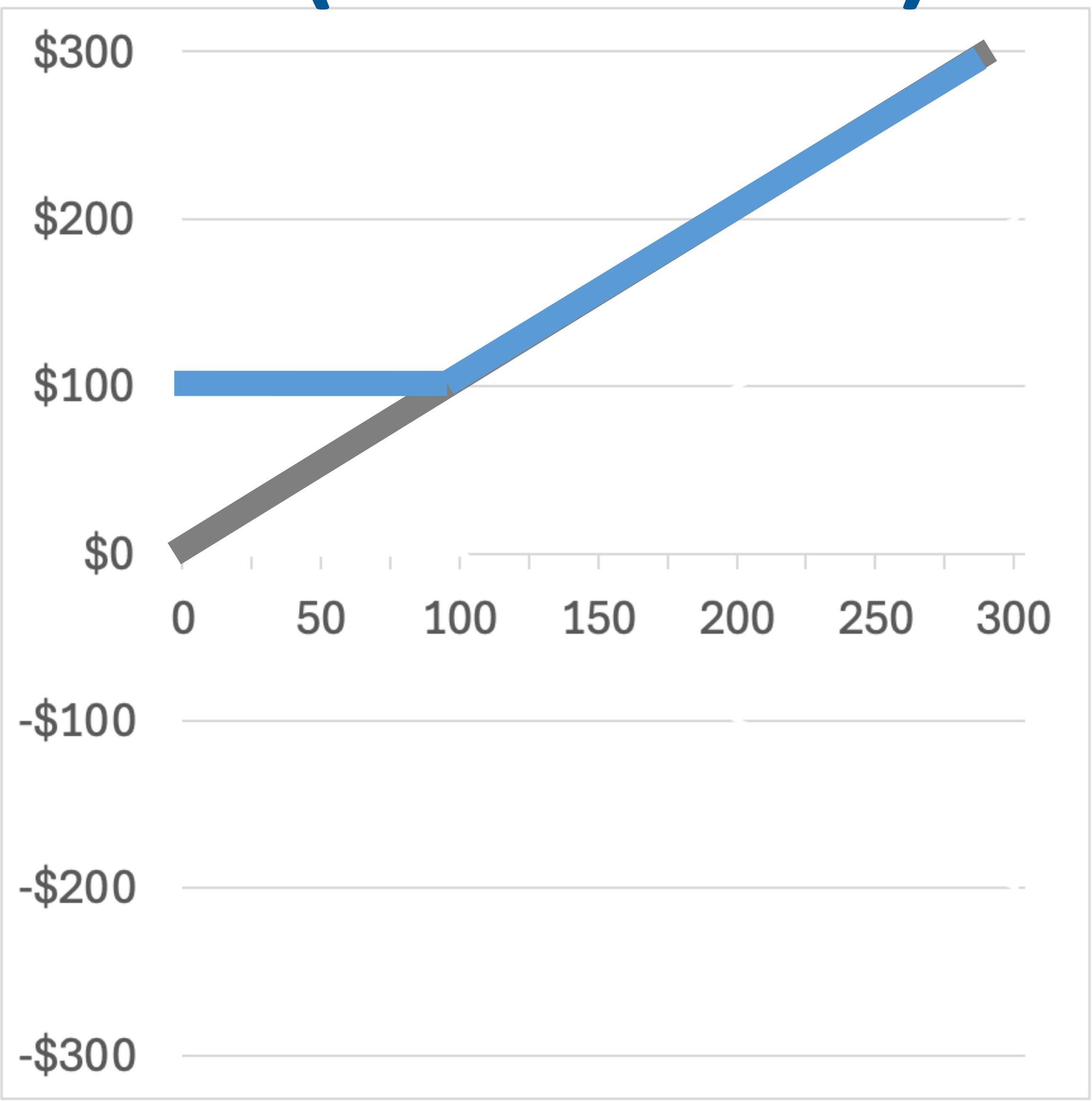

Value of common stock at expiration ($)

Put-Call Parity

$$C + PV(K) = P + S$$

- C and P are the value of call and put options.
- PV(K) is the present value of the strike price.
- S is the share price.

The payoffs at time T are the same on both sides,

irrespective of S_T (the stock price at expiration).

Why? If there were differences, there is an arbitrage opportunity.

Valuing Options Before Maturity

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- So far, only valued options *at maturity*. But what are they worth before maturity?

Valuing Options Before Maturity

- So far, only valued options *at maturity*. But what are they worth before maturity?
- Their price will change depending on the stock price over the life of the option.
  - What is the opportunity cost given the risk-free rate?
  - How volatile is the stock price?
  - How long left until the option matures?
- Expected value of a call at maturity: $E[\max(S_T - K, 0)]$
- Expected value of a call before maturity: $\mathrm{PV}(E[\max(S_T - K, 0)])$

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Black-Scholes Model is a way to price options (brief introduction – you are not expected to know this math)

- Expected value of a call (at/before maturity):

$$\mathrm{C} = \mathrm{PV}(E[\max(S_T - K, 0)])$$

$$\mathrm{C} = e^{-r(T-t)}(E[\max(S_T - K, 0)])$$

$$\mathbf{C} = \boldsymbol{S\Phi}[\boldsymbol{d_1}] - \boldsymbol{Ke^{-r(T-t)}\Phi}[\boldsymbol{d_2}]$$

And price of a put (at/before maturity):

$$\mathbf{P} = \boldsymbol{Ke^{-r(T-t)}\Phi}[-\boldsymbol{d_2}] - \boldsymbol{S\Phi}[-\boldsymbol{d_1}]$$

$\boldsymbol{\Phi}$ is standard normal cumulative distribution function. S is current price, K is strike price, r is risk-free rate.

d1 is $\frac{\ln\left(\frac{S}{K}\right)+\left(r+\frac{\sigma^2}{2}(T-t)\right)}{\sigma\sqrt{T-t}}$ and d2 is $\frac{\ln\left(\frac{S}{K}\right)-\left(r+\frac{\sigma^2}{2}(T-t)\right)}{\sigma\sqrt{T-t}}$

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Black-Scholes Model Assumptions

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  - Stock will not pay dividends until after the expiration date
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  - Volatility is constant
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  - Stock will not pay dividends until after the expiration date
  - Interest rate is constant
  - Volatility is constant
  - No extreme changes in prices (e.g., no jumps)
- Relaxing these assumptions can require hard math
- This is why Wall Street hires people with advanced degrees in maths, physics etc.

Other Derivatives

Using Options and Derivatives

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Financial options are not only used for hedging.

- **"Speculators"** use options to make leveraged bets.

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- Futures are a type of derivative, but not an option (because you are required to buy/sell rather than having the option to do so).
- Traded through through futures exchange.

Examples of Futures (WSJ)

AGRICULTURE	LAST	CHG	% CHG
Corn	492.50	2.25	0.46%
Wheat	557.00	-5.25	-0.93%
Soybeans	1044.25	-3.50	-0.33%
Soybean Oil	48.25	-0.09	-0.19%
Soybean Meal	301.6	-1.5	-0.49%
Live Cattle	203.950	-0.125	-0.06%

CURRENCIES	LAST	CHG	% CHG
Euro	1.1527	0.0114	0.99%
Yen	0.7140	0.0073	1.03%
Pound	1.3385	0.0114	0.86%
Australian Dollar	0.6412	0.0017	0.26%
Canadian Dollar	0.7262	0.0016	0.21%

INTEREST RATES	LAST	CHG	% CHG
Eurodollar	94.4900	-0.0075	-0.01%
U.S. 2 Year	103.922	0.129	0.12%
U.S. 5 Year	108.492	0.078	0.07%
U.S. 10 Year	111.016	-0.156	-0.14%
U.S. 30 Year	113.438	-1.125	-0.98%

Futures

11:22 AM ET 04/21/25

ENERGY	LAST	CHG	% CHG
Crude Oil	63.00	-1.68	-2.60%
Brent Crude	66.18	-1.78	-2.62%
Natural Gas	3.091	-0.154	-4.75%
RBOB Gasoline	2.0570	-0.0418	-1.99%
Heating Oil	2.0702	-0.0421	-1.99%

METALS	LAST	CHG	% CHG
Gold	3430.30	101.90	3.06%
Silver	32.645	0.175	0.54%
Copper	4.7440	0.0050	0.11%

INDEXES	LAST	CHG	% CHG
E-Mini Dow	38367	-962	-2.45%
E-Mini S&P 500	5180.25	-132.50	-2.49%
E-Mini Nasdaq 100	17889.25	-491.50	-2.67%

Corn Futures Example (CME)

CORN FUTURES - CONTRACT SPECS

CONTRACT UNIT	5,000 bushels
PRICE QUOTATION	U.S. cents per bushel
TRADING HOURS	**CME Globex:** Sunday - Friday, 7:00 p.m. - 7:45 a.m. CT and Monday - Friday, 8:30 a.m. - 1:20 p.m. CT TAS: Sunday - Friday 7:00 p.m. - 7:45 a.m. and Monday - Friday 8:30 a.m. - 1:15 p.m. CT **CME ClearPort:** Sunday 5:00 p.m. - Friday 5:45 p.m. CT with no reporting Monday - Thursday from 5:45 p.m. - 6:00 p.m. CT
MINIMUM PRICE FLUCTUATION	1/4 of one cent (0.0025) per bushel = $12.50 TAS: Zero or +/- 4 ticks in the minimum tick increment of the outright
PRODUCT CODE	**CME Globex:** ZC **CME ClearPort:** C **Clearing:** C **TAS:** ZCT
LISTED CONTRACTS	9 monthly contracts of Mar, May, Sep and 8 monthly contracts of Jul and Dec listed annually after the termination of trading in the December contract of the current year.
SETTLEMENT METHOD	Deliverable

https://www.cmegroup.com/markets/agriculture/grains/corn.contractSpecs.html

Forwards

Forwards are Bespoke Future Contracts

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- Forwards are tailored in their asset, amount and delivery date.
- Forwards settled on the settlement date (not marking-to-market). This means that if the price has varied significantly since the contract was made, there is a risk that the other party defaults on the contract.
- “Taking delivery” when receive the deliverable instrument forward based on.
- Forwards are a type of derivative, but not an option (because you are required to buy/sell rather than having the option to do so).

Swaps

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- Common swaps include:
  - Foreign exchange swaps (swapping cash flows based on exchange rates)
  - Interest rate swaps (swapping cash flows based on interest rates)
  - Credit default swaps (CDS, swapping cash flows based on default rates)
- Swaps are bespoke and traded over-the-counter (OTC).

Famous Derivative Trading Examples

When Derivatives Trading Goes Wrong: LTCM

Long-Term Capital Management (LTCM), 1998

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- NAV ~$4bn. Swaps ~$1.25tr (~5% of global market!)
- Russian sovereign default in 1998 triggered 'flight to safety'
- Strategies they thought were diversified were not – unhedged to liquidity and as had illiquid positions they couldn't exit!
- Lost ~$4.6 bn in few months, and Federal Reserve arranged other banks to bailout.

When Derivatives Trading Goes Wrong: Barings

“Rogue Trader” Nick Leeson at Barings Bank, 1995

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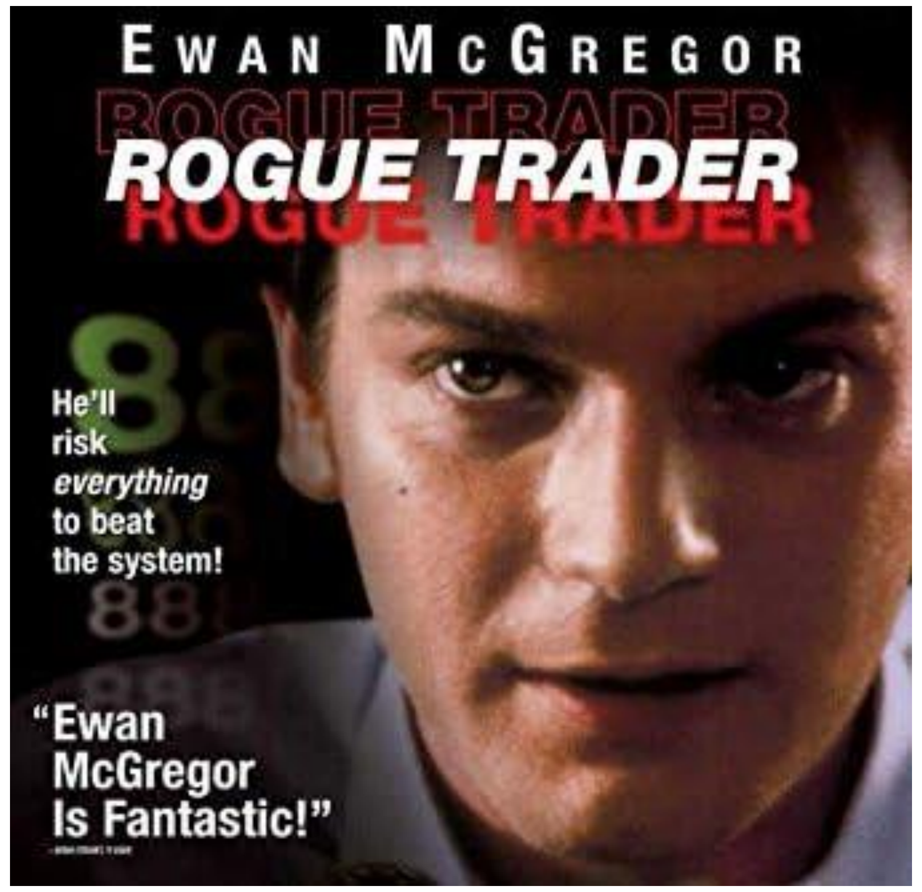

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- Barings Bank was a British (merchant) bank dating back to 1762!
- Nick Leeson traded derivatives for them in Singapore

- Was supposed to be buying on one market and selling on another market.
- "Short straddle" strategy (selling both call and put options) on Singapore and Tokyo exchanges betting little change overnight. Strategy produces payoffs if little volatility.

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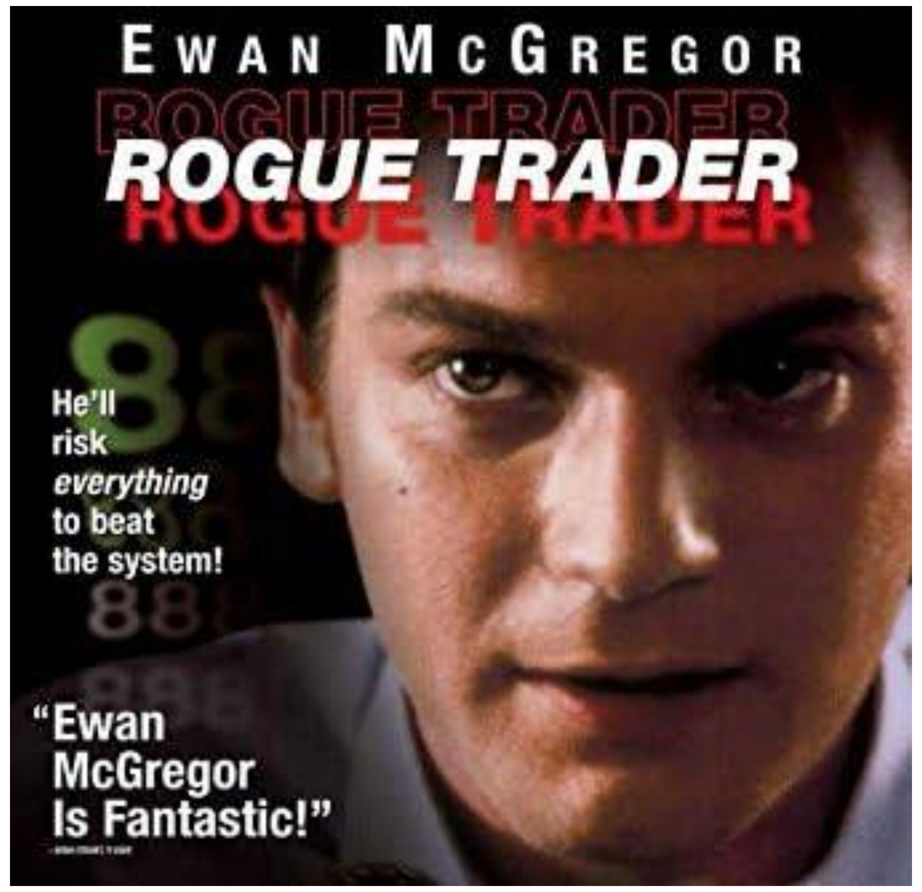

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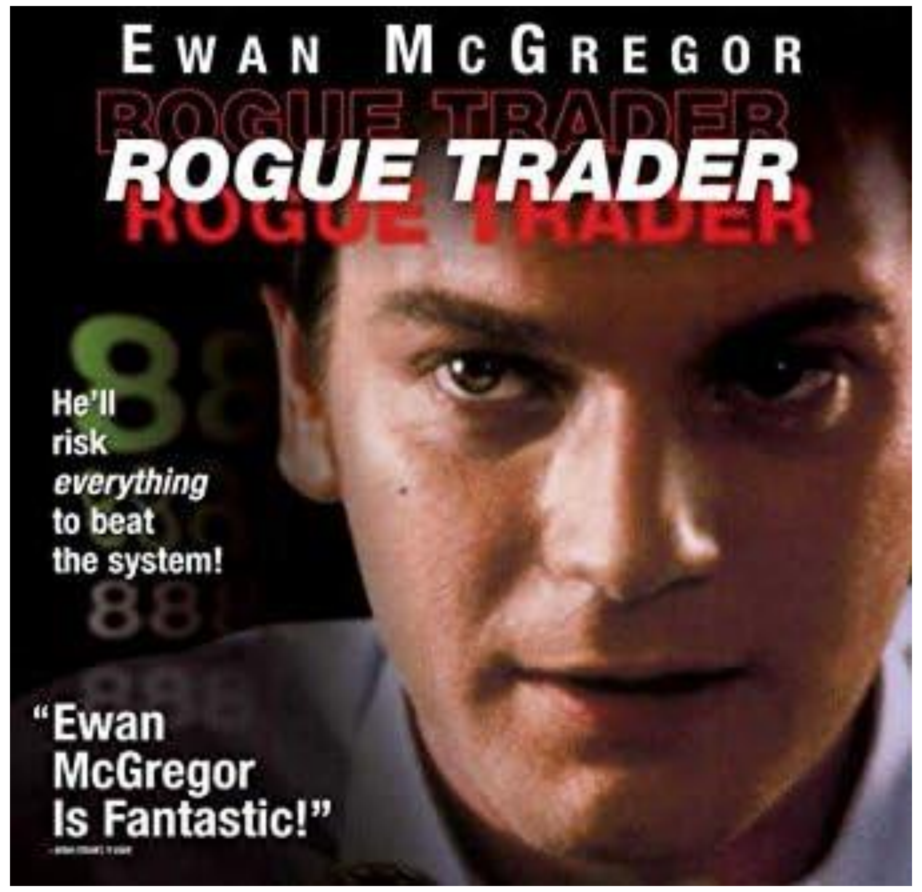

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- Earthquake struck Kobe. Huge losses in Asian markets.
- Leeson then made increasingly risky (unauthorized) trades to try to recover losses.
- Lost ~$1.4bn in Singapore Exchange
- Barings Bank went bankrupt (his losses were twice Baring's trading capital) and Leeson went to prison for fraud.

Week 10 Examples

Q1. Trader GK holds a 100 one-year European call option contract that enables purchasing 100 shares in Bunny Inc, with an exercise price of $150. The option costs $200. What is the total payoff to Trader GK if Bunny Inc's share price at expiration is
(a) $200 (b) $50?

Q2. Trader GK holds a 100 one-year European put option contract that enables selling 100 shares in Bunny Inc for $70. The option costs $200. What is the total payoff to Trader GK if Bunny Inc's share price at expiration is
(a) $150 (b) $50?

Q3. Calculate the value of portfolio with a put (K=$30) and the underlying stock if the stock price at expiration is (a) $20, (b) $30, © $70

	Stock Price = $20	**Stock Price = $30**	**Stock Price = $70**
Stock Value			
Put Value			
Portfolio Value			

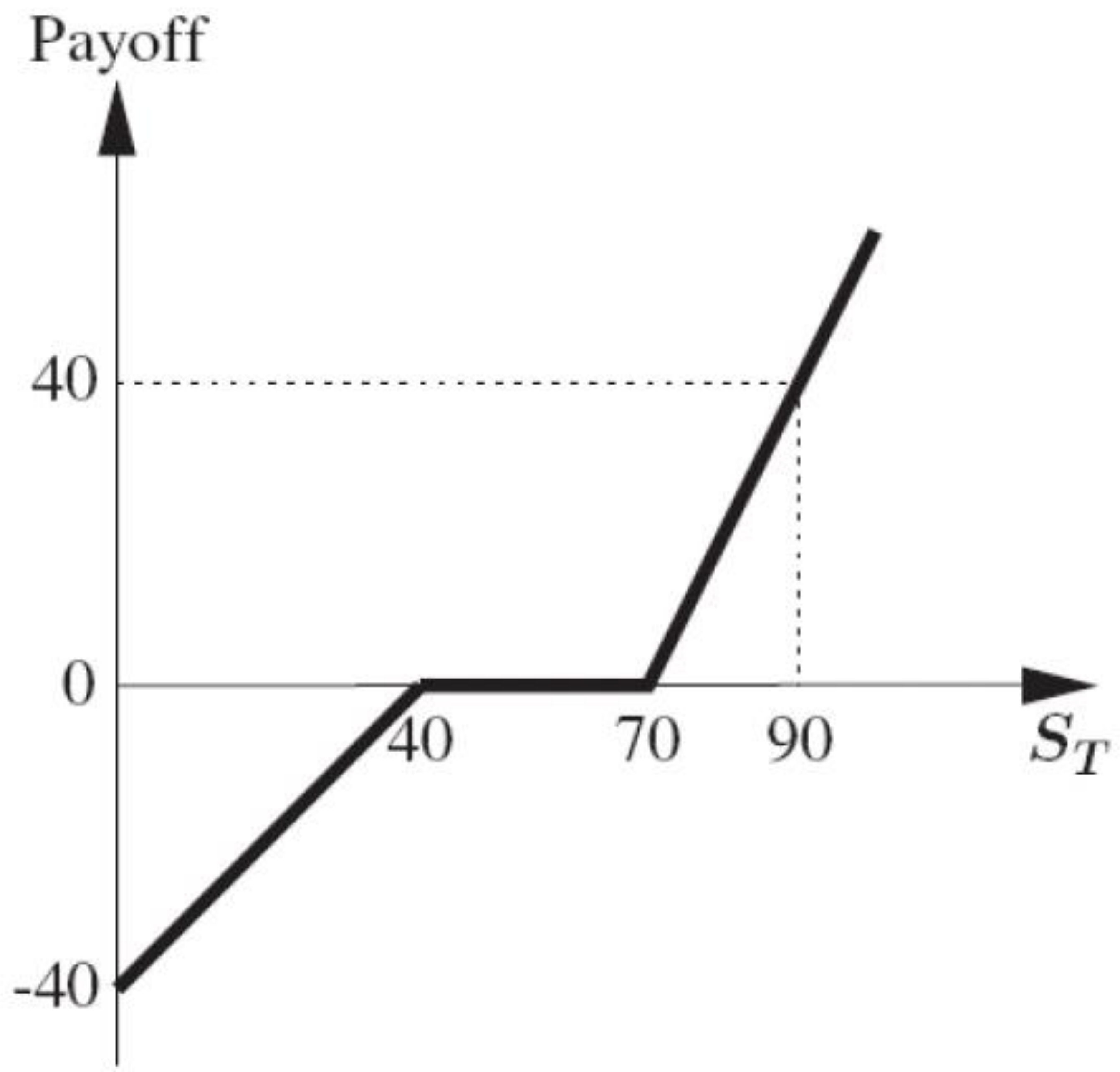
Q4. Both the exercise price of the call and put options on Hot Cross Buns Limited are $55. Both options are European, so they cannot be exercised prior to expiration. The expiration date is one year from today. The current stock price is $44. After the expiration date, the stock will either be $34 or $58. What is payoff of (a) buying the call and investing in bonds at an amount equal to PV(K) and (b) buying both the put and the stock?

(a)	Stock Price = $34	Stock Price = $58
Call Value		
Bond Value		
Portfolio Value		

(b)	Stock Price = $34	Stock Price = $58
Put Value		
Stock Value		
Portfolio Value		

Q5. Carrot Company Is currently trading for $35 per share. In exactly one year it will be worth either $47 or $30 with equal probabilities. The risk-free rate is 6%.
(a) What is the expected return on Carrot Company?
(b) What is the price of a K=40 European call option that expires in exactly one year?
(c) What is the expected return on the option?

Q6. Benedict is a trader who wants to replicate the payoffs as shown below. He can only trade in European call options and European put options on a stock. These can have any strike price, but have the same maturity T. What portfolio of calls and/or puts would replicate the payoff at maturity?



Q7. Benedict is a trader who wants to replicate the payoffs as shown below. He can only trade in European call options and European put options on a stock. These can have any strike price, but have the same maturity T. What portfolio of calls and/or puts would replicate the payoff at maturity?